Adaptive Domain-Adversarial Few-Shot Learning for Cross-Domain Hyperspectral Image Classification

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***Abstract*— The process of annotating hyperspectral image (HSI) data is characterized by its time-consuming and labor-intensive nature. To address this challenge, researchers often employ a meta-learning paradigm known as few-shot learning (FSL), which leverages source domains containing a substantial number of labeled samples to assist in the classification of target domains with limited labeled samples. Many existing FSL methods rely on a conditional domain-adversarial strategy to mitigate the domain shift between the source and target domains. However, these methods overlook the fact that the degrees of conditional distribution discrepancies between the two domains can vary significantly across different classes, leading to suboptimal conditional distribution alignment. To address this problem, we propose a framework called adaptive domain-adversarial FSL (ADAFSL). Overall, the proposed ADAFSL employs an adaptive strategy that assigns varying weights to the conditional adversarial losses for different classes based on their respective degrees of discrepancies, thereby achieving global conditional distribution alignment. Specifically, a local alignment score map is constructed by measuring the similarity between labeled and unlabeled samples using both Euclidean and class-covariance metrics. This map is then multiplied with the conditional adversarial loss map, thus allocating more emphasis to the classes exhibiting greater discrepancies between the two domains. Moreover, to enhance cross-domain FSL, we design a multiscale spectral–spatial feature extraction (MSFE) module that incorporates cascaded multiscale dilated convolutions. Experimental results on four public HSI datasets demonstrate that the proposed ADAFSL outperforms other state-of-the-art methods. The source code of this method can be found at https://github.com/JieW-ww/ADAFSL.**

***Index Terms*— Conditional adversarial domain adaptation, cross-domain few-shot learning (FSL), meta-learning, multiscale feature extraction.**

I. Introduction

COMPARED to conventional color images, hyperspectral images (HSIs) encompass a large number of continuous wavebands, offering a wealth of detailed spectral–spatial information [1]. This characteristic enables precise identification of ground objects, leading to widespread applications in military target detection, medical diagnosis, and precision agriculture [2], [3], [4]. With the advent of deep learning, ground object classification using HSIs has witnessed remarkable advancements, prompting the development of numerous deep-learning models specifically tailored for this task [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21]. Deep-learning methods exhibit exceptional feature extraction and learning capabilities, automatically deriving complex features from HSIs through the construction of deep network architectures and large-scale data training strategies, thereby achieving accurate classification results.

Despite the remarkable success of deep-learning HSI classification, training deep-learning models often necessitates a substantial amount of labeled data. However, the process of annotating HSIs to obtain a sufficient number of labeled samples is known to be time-consuming and laborious. The high cost associated with manual labeling significantly limits the learning potential of current deep-learning models. Moreover, practical scenarios often present situations where the HSI dataset to be classified contains few labeled samples (referred to as the target domain), while another unrelated HSI dataset contains an ample number of labeled samples (referred to as the source domain). To address this challenge, cross-domain learning has emerged as a methodology that leverages sample information from the source domain to enhance the classification performance in the target domain and to reduce the demand for training samples in the target domain. However, the source and target domains typically exhibit different data distributions, necessitating the development of various methods for cross-domain learning. These methods can be broadly categorized into two types: those based on domain adaptation [22], [23] and those based on fine-tuning [24], [25].

In the domain-adaptation method, the central objective is to focus on feature-level domain adaptation, which involves aligning the feature space of the source and target domains directly, thereby reducing their distribution discrepancy and achieving enhanced generalization performance on the target domain. Jia et al. [26] proposed a gradient feature-oriented 3-D domain adaptation for HSI classification. This approach employs a 3-D domain-adaptation method that jointly utilizes second- and fourth-order statistical descriptors in the spatial–spectral dimensions, followed by 3-D Sobel–Gabor gradient (3DSGG) and 3-D derivative-of-Gaussian–Gabor (3DDGG) feature extraction. In a similar vein, Ye et al. [27] introduced a dictionary learning-based feature-level domain

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adaptation for cross-scene HSI classification. This method aligns the spectral distributions between the source and target scenes by projecting their spectral features into a shared low-dimensional embedding space through multitask dictionary learning. It is important to note that these methods often assume that the source and target domains have similar class labels, implying consistency between the classes in the source and target domains. Consequently, these methods may encounter challenges when precisely classifying samples from new classes that exist solely in the target domain but not in the source domain.

Regarding the fine-tuning method, the initial step involves transferring the first layer weight of the pretraining network to the target model, followed by fine-tuning the top layer of the pretraining network using the backpropagation algorithm. Mei et al. [24] proposed a five-layer convolutional neural network (CNN) (C-CNN) trained in the sensor-specific source domain. They then utilized the pretrained fully connected layers from this C-CNN to construct a companion feature-learning CNN (FL-CNN) in the target domain, employing the same HSI sensor as the source domain. On a similar note, Yang et al. [25] introduced a two-branch CNN to extract joint spectral–spatial features from HSIs. In this method, the pretrained low and mid-layers of the network from the source domain remain fixed, while its top layers are trained with limited training samples from the target domain. It is important to emphasize that the classes in the source and target domains need not be consistent, as this approach relies not only on extensive pretraining with source-domain data, but also on the availability of sufficient labeled data for new classes in the target domain. However, it is worth noting that the fine-tuning method is inadequate in handling situations where significant data distribution shifts occur between the two domains, and when a large number of labeled samples are required in the target domain.

The scarcity of labeled samples in HSIs is primarily due to the high cost and labor involved in manual annotation, significantly restricting the practical application of deep-learning approaches. To address this limitation, researchers have introduced few-shot learning (FSL) methods, a form of meta-learning [28], also known as "learning how to learn." FSL methods are designed to handle the specific challenge of limited labeled samples. In an FSL scenario, the model learns general information (meta-knowledge) from the labeled samples of numerous classes in the source domain. This learned knowledge can then assist in predicting target classes when only a few labeled samples are available in the target domain. Liu et al. [29] proposed a deep FSL (DFSL) method based on [30], specifically tailored for classifying HSIs with limited labeled data. DFSL learns the metric space from the source dataset through a deep residual network based on a 3-D CNN. During the testing phase, DFSL utilizes SVM [31] and K-nearest neighbor (K-NN) [32] methods to classify unlabeled data from the test dataset. Furthermore, Gao et al. [33] introduced a relation network (RN-FSC) model for HSI few-shot classification, where a small number of labeled target-class data are used to fine-tune the model, achieving promising classification results.

The aforementioned FSL methods can enhance classification accuracy with a limited number of labeled test samples. However, these methods typically assume that the conditional distributions of both the source and target domains are identical. In reality, the distributions of source and target domains often differ, leading to the domain-shift problem. To address this challenge, Li et al. [34] proposed a deep cross-domain FSL (DCFSL) approach. This method adopts the meta-learning paradigm for cross-domain classification with few samples and integrates a conditional domain-adversarial strategy to tackle the domain-shift issue. On a similar note, Xi et al. [35] introduced FSL with class-covariance metric (CMFSL). This method learns domain-invariant class representations by conducting interactive training between the base and novel classes and employing the Mahalanobis distance measurement instead of an extra classifier. Zhang et al. [36] introduced a graph information aggregation cross-domain FSL (Gia-CFSL) for HSI Classification. This method employs a strategy of graph neural network information propagation and graph alignment. It considers the nonlocal spatial information with strong correspondence in both the feature and distribution levels to perform cross-domain graph alignment, thereby effectively mitigating domain shift issues.

Although the aforementioned FSL methods show promise in partially alleviating the domain shift between the source and target domains, they do have certain limitations. First, when adopting a conditional domain-adversarial strategy to achieve conditional distribution alignment, these methods fail to consider the varying degrees of dissimilarity in the conditional distributions of different classes between the two domains. As a consequence, the alignment may not be optimal, and negative transfer could be introduced to the aligned features. Second, using residual-connected CNNs for extracting spatial–spectral features from HSIs demands more computational resources and does not guarantee the extraction of fine-grained features. These limitations call for the development of more sophisticated methods to enhance domain alignment and feature extraction in the context of FSL for HSI classification.

To address the limitations mentioned above, we present a novel framework called adaptive domain-adversarial FSL (ADAFSL) tailored for HSI classification. ADAFSL leverages an adaptive conditional domain-adversarial strategy, employing Euclidean and class-covariance metrics to assess the similarity between labeled and unlabeled samples, thereby generating a local alignment score map. This score map is then multiplied with the conditional adversarial loss map obtained from the domain discriminator, allowing for class-specific weighting of samples in the source and target domains. This adaptive strategy empowers the model to iteratively focus on classes with significant distribution disparities between the two domains, effectively improving cross-domain performance and mitigating domain shift effects. Furthermore, ADAFSL introduces a feature extraction network equipped with convolutions of different scales and receptive fields, enabling the extraction of rich and diverse spectral–spatial features from HSIs. The core component of this network is the multiscale spectral–spatial feature extraction (MSFE) module, which

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employs cascaded dilated convolutions to facilitate feature reuse without introducing additional parameters. Moreover, the MSFE module effectively extracts fine-grained features, enhancing the characterization of the internal structure of HSIs. By applying MSFE modules in different domains, the proposed network promotes feature alignment between the source and target domains, thereby bolstering cross-domain classification performance. The ADAFSL framework overcomes the shortcomings of existing methods by integrating an adaptive domain-adversarial strategy and a sophisticated feature extraction network, making it a promising and robust approach for HSI classification in the presence of limited labeled samples and domain shift challenges.

The main contributions can be summarized as follows.

1) A framework called ADAFSL is proposed for HSI classification, aiming to globally align the conditional distributions of all corresponding classes between the two domains and extract domain-invariant spectral–spatial features. By transferring knowledge from the source domain, which has sufficient labeled samples, to the target domain with limited labeled samples, ADAFSL enables accurate cross-domain classification.
2) To achieve effective alignment, we introduce an adaptive conditional domain-adversarial strategy. This strategy dynamically weights the adversarial loss of each feature based on the alignment level of each class. Prioritizing the conditional distributions of classes with higher mismatches between the source and target domains effectively mitigates the negative impact of feature mismatch resulting from domain shift.
3) To learn a comprehensive spatial–spectral metric space, we propose a multiscale feature extractor. This feature extractor includes the MSFE module, which captures fine-grained spatial–spectral features at different scales of the receptive field. The module then concatenates features extracted by each layer of convolution, promoting feature reuse and reducing the number of parameters through cascaded dilated convolutions. These advancements enhance the capability of the proposed ADAFSL framework in addressing the challenges of FSL and domain shift in HSI classification.

The remainder of this article is organized as follows. Section II describes in detail the relevant theories of cross-domain FSL and conditional domain-adversarial strategy. Section III describes the proposed ADAFSL. Section IV discusses the experimental results and the effect of parameters in the ADAFSL design. In Section V, we conclude the article.

II. Related Work

A. Cross-Domain FSL

Meta-learning, as a noteworthy methodology to tackle few-shot classification, extracts transferable knowledge aiding insufficient labeled samples to adequately train models in units of episodes and reflect inter- and intraclass variations [37]. Each episode can be divided into support and query sets, and the support set vector is used to train a "center" for each class, which is called "learn to learn." The classes of the samples in the query set are determined by their distance from the "center." In the context of limited labels, the support set is selected in a C-way-K-shot manner, where the support set contains K labeled samples for each of C classes. In a C-way-K-shot classification task, the objective is to learn a C-class classification model using the labeled training data of each class limited to K samples. During the training phase, FSL methods derive general patterns from labeled samples of a wide range of classes in the source domain. Consequently, when encountering a new and unknown target class, these FSL methods leverage the meta-knowledge acquired from the source domain to perform accurate classification. Notably, the classes between the source and target domains do not necessarily have to be the same, as the general knowledge learned from the source domain can significantly aid in feature extraction, class classification, and generalization reasoning by the model. This property allows FSL methods to achieve promising results even when faced with the challenges posed by the scarcity of labeled samples and the domain shift between the source and target domains.

In the setting of cross-domain FSL, we have two datasets: $\mathbf{D}_S$, the source-domain dataset containing sufficient labeled samples for c_s classes, and $\mathbf{D}_T$, the target-domain dataset with a limited number of labeled samples for c_t classes. To ensure effective learning of meta-knowledge in the training stage [34], it is essential that c_s is greater than c_t, as this facilitates the extraction of comprehensive meta-information from the source domain. The target-domain dataset can be further divided into two subsets: the few-shot dataset $\mathbf{D}_f$, which contains labeled data, and the test dataset $\mathbf{D}_t$, comprising unlabeled data. In our proposed method, we utilize the source dataset and the few-shot dataset for training the embedded feature extractor, evaluating its generalization ability on the test dataset. During the FSL and training process, a small number of samples with episodes are used to simulate the FSL task. Each episode randomly selects samples from either the source- or target-domain dataset, creating specific source- and target-domain training tasks. The FSL is accomplished by iteratively training these tasks from the source and target domains alternately.

Taking target domain training as an example, as shown in Fig. 1, m samples per class are randomly selected from target domain to form a support set $\mathbf{S}_t = \{(\mathbf{x}_i, y_i)\}_{i=1}^{c_t \times m}$, and then n samples are randomly selected to form a query set $\mathbf{Q}_t = \left\{\left(\mathbf{x}_j, y_j\right)\right\}_{j=1}^{c_t \times n}$. The class distribution of the sample $\mathbf{x}_j$ in the the query set $\mathbf{Q}_t$ is calculated as

$$P\left(y_j = k \mid \mathbf{x}_j \in \mathbf{Q}_t\right) = \frac{\exp\left(-d\left(f_\varphi\left(\mathbf{x}_j\right), \mathbf{c}_k\right)\right)}{\sum_{k=1}^{c_t} \exp\left(-d\left(f_\varphi\left(\mathbf{x}_j\right), \mathbf{c}_k\right)\right)} \quad (1)$$

where $d(\cdot, \cdot)$ represents the Euclidean metric, $\mathbf{c}_k$ is the embedded feature of the kth class in the support set, c_t is the number of classes in each set, $f\varphi(\cdot)$ represents the embedded feature extraction network with parameter φ, $\mathbf{x}_j$ is the query set sample, and y_j is the label of $\mathbf{x}_j$.

During the training stage, the support set is used to train the model, and the query set is used to calculate the loss function, which, in turn, updates the model parameters. The FSL training process in the source domain is analogous to that in the target domain. When the labeled-sample size is small, it becomes challenging to generate sufficient support and query

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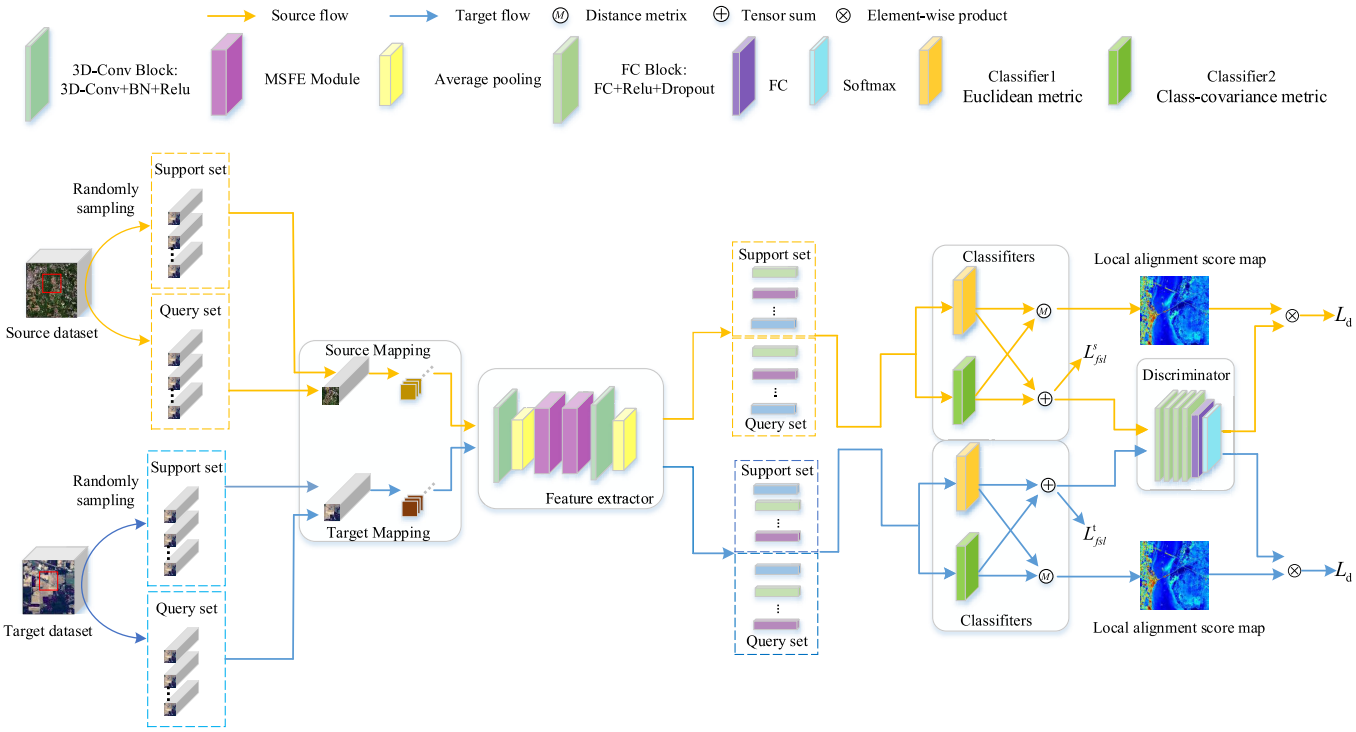


Fig. 1. Overall framework of the proposed ADAFSL in the training phase.

data. To address this, data augmentation techniques such as adding random Gaussian noise are commonly employed to expand the number of samples [34].

B. Conditional Domain-Adversarial Strategy

The domain-adversarial strategy [38], [39] combines adversarial learning with domain adaptation, resembling the concept of generative adversarial networks (GANs) [40]. In this approach, a domain discriminator is trained to minimize the domain classification error, distinguishing the source domain from the target domain. Simultaneously, the classification model learns transferable knowledge that the domain discriminator cannot differentiate. However, the domain-adversarial strategy may encounter challenges in accurately representing features when the data distribution contains multimodal structures, leading to a failure in achieving discriminant alignment due to model mismatch. This risk arises, first, from the inherent equilibrium challenge of adversarial learning. Even if the discriminators are unable to distinguish between the two distributions, it does not necessarily guarantee that the distributions are entirely similar. Second, the error probability of the domain discriminator increases when the discriminant information is uncertain [41].

The limitations of the domain-adversarial strategy should be given careful consideration, particularly when dealing with cross-domain learning tasks involving multimodal data distributions. These limitations may affect the effectiveness and accuracy of feature representation and discriminant alignment. To address these challenges, the conditional domain-adversarial strategy [42], [43] has been developed to handle datasets with high variability and multimodal distributions, providing solutions to these issues. In the conditional-adversarial domain-adaptation network proposed in [41], the discriminant information conveyed by the classifier is utilized to assist adversarial adaptation. The core of this approach lies in a cross-entropy conditional domain discriminator, which leverages domain-specific feature representation and classifier prediction. Moreover, the domain discriminator is set as an uncertain condition based on classifier prediction, with priority given to easy-to-transfer samples.

In the conditional domain-adversarial strategy, the discriminant information conveyed by the classifier prediction G potentially reveals the multimodal structure, which can serve as a condition for feature representation F. Through conditional action, the domain-invariant feature representation F and classifier prediction G can be jointly modeled. The conditional adversarial domain-adaptation network is formulated as a minimax optimization problem with two competing error terms: 1) $E(F, G)$, which is minimized on the classifier G to ensure a low classification risk on the source domain and 2) $E(D, F, G)$, as shown in (2), is minimized on the domain discriminator D to distinguish between the source and target domains, and maximized on the feature representation F and classifier prediction G to ensure a domain-invariant feature extractor and classifier

$$\min_{D}\max_{F,G} E(D, F, G) = -E_{x_i^s \sim P_s(x)} \log\left[D\left(f_i^s, g_i^s\right)\right] - E_{x_j' \sim P_t(x)} \log\left[1 - D\left(f_j^t, g_j^t\right)\right]. \tag{2}$$

III. Methodology

A. Proposed ADAFSL

As depicted in Fig. 1, the proposed ADAFSL framework comprises two mapping layers to ensure that input data from

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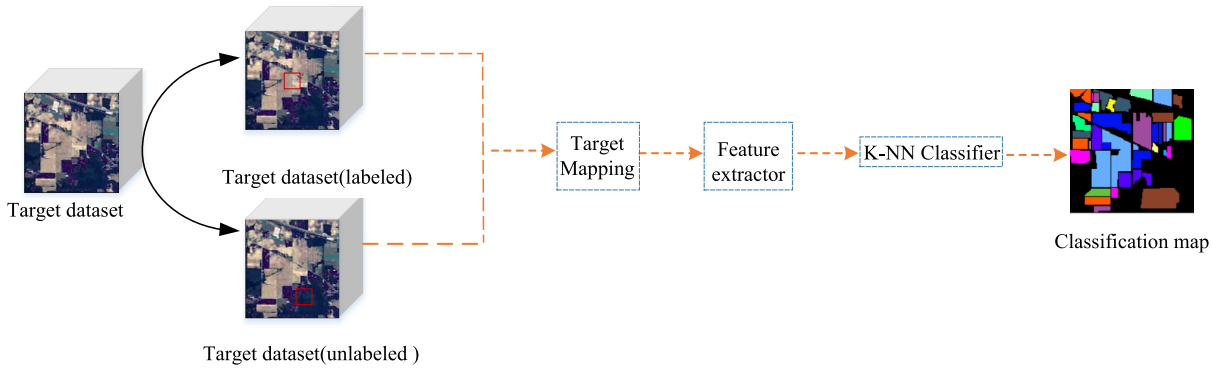


Fig. 2. Overall framework of the proposed ADAFSL in the testing phase.

the source and target domains have the same dimensions before entering the embedded multiscale feature extractor. The feature extractor then maps the input data samples to a domain-invariant metric space. The adaptive domain discriminator is responsible for achieving domain distribution alignment at the class level. The ADAFSL framework involves two types of FSLs: source-domain FSL and target-domain FSL. After embedding the input data from both domains into the spectral–spatial embedded metric space, FSL is performed by calculating the distance between labeled and unlabeled samples. Finally, the self-adaptive domain discriminator is employed to realize domain distribution alignment at the class level, effectively reducing the domain shift between the source and target domains. For a detailed description of the training process, refer to Algorithm 1.

During the FSL training process on the source domain, two different distance metrics are utilized to calculate the distance between labeled and unlabeled samples: the Euclidean-based method and the class-covariance-based method. The class-covariance-based method offers more flexibility in decision boundary construction compared to the Euclidean-based method. Two classifiers based on the two metrics produce predictions denoted as $\mathbf{p}^1$ and $\mathbf{p}^2$, respectively. These predictions are then combined through additive operations to generate integrated predictions denoted as $\mathbf{p}$. Subsequently, the predictions $\mathbf{p}$ are used to compute the FSL loss under the supervision of the true label. Simultaneously, the prediction results $\mathbf{p}^1$ and $\mathbf{p}^2$ produced by the two classifiers are computed by cosine distance metric for their local alignment score map, and this result is multiplied element-by-element with the conditional adversarial loss map obtained by the integrated predictions $\mathbf{p}$ passing domain discriminator to subsequently compute the adaptive conditional domain-adversarial loss. The combined loss from both the FSL and the adaptive conditional domain-adversarial components contributes to the training process, enabling the model to learn and adapt effectively across the source domain with sufficient labeled samples and the target domain with limited labeled samples. This result is then multiplied element-by-element with the adversarial loss map obtained by the p-passing domain discriminator to subsequently compute the adaptive conditional domain adversarial loss.

Based on the negative log probability of query sample $\mathbf{x}$ and its true class label k, the FSL loss for all query samples

Algorithm 1 Pseudocode for the Training Process of the Proposed ADAFSL

Input: $\mathbf{S}_t$, $\mathbf{Q}_t$ of $\mathbf{D}_T$, $\mathbf{S}_s$, $\mathbf{Q}_s$ of $\mathbf{D}_S$, the number of training episodes,
hyperparameters λ_{adv} and λ_{local}.
Output: The classification accuracy of each class of the target dataset.
1:**Few-Shot Learning**
2:**for** episode = 1: episodes **do**
3: Randomly selected $\mathbf{S}_t$, $\mathbf{Q}_t$ from $\mathbf{D}_T$, $\mathbf{S}_s$, $\mathbf{Q}_s$ from $\mathbf{D}_S$;
4: Feature extraction via the network described in Table I;
5: Calculate L^s_{fsl}, L^t_{fsl} by Eqs.(3) and (4);
6: $Loss_{\text{fsl}} = L^s_{\text{fsl}} + L^t_{\text{fsl}}$;
7: $Loss_{\text{fsl}}$.backward()
8:**end for**
9:**Adaptive Class-Level Conditional Domain Discriminator**
10:**for** episode = 1: episodes **do**
11: Calculate $\mathbf{p}^1$, $\mathbf{p}^2$ by Classifiers;
12: Calculate $M(\mathbf{p}^1, \mathbf{p}^2)$ by Eq.(7);
13: Calculate adversarial loss by Discriminator;
14: Calculate L_d by Eq.(6);
15: $Loss = L_{\text{fsl}} + \lambda_{adv} L_d$
16: $Loss$.backward()
17:**end for**

in the source domain can be expressed as

$$L^s_{\text{fsl}} = E_{\mathbf{S}_s,\mathbf{Q}_s} \sum_{(\mathbf{x},y)\in\mathbf{Q}_s} -\log(\mathbf{p}(y = k \mid \mathbf{x})) \tag{3}$$

where $\mathbf{S}_s$ and $\mathbf{Q}_s$ are the support and query sets from the source-domain dataset, respectively. The target domain performs the same way as the source domain, so the FSL loss in the target domain is calculated as

$$L^t_{\text{fsl}} = E_{\mathbf{S}_t,\mathbf{Q}_t} \sum_{(\mathbf{x},y)\in\mathbf{Q}_t} -\log(\mathbf{p}(y = k \mid \mathbf{x})) \tag{4}$$

where $\mathbf{S}_t$ and $\mathbf{Q}_t$ represent the support and query sets from the target-domain dataset, respectively. After incorporating the adaptive conditional domain-adversarial loss, the total loss functions for the source and target domains are defined as follows:

$$L^s_{\text{total}} = L^s_{\text{fsl}} + \lambda_{\text{adv}} L_d$$

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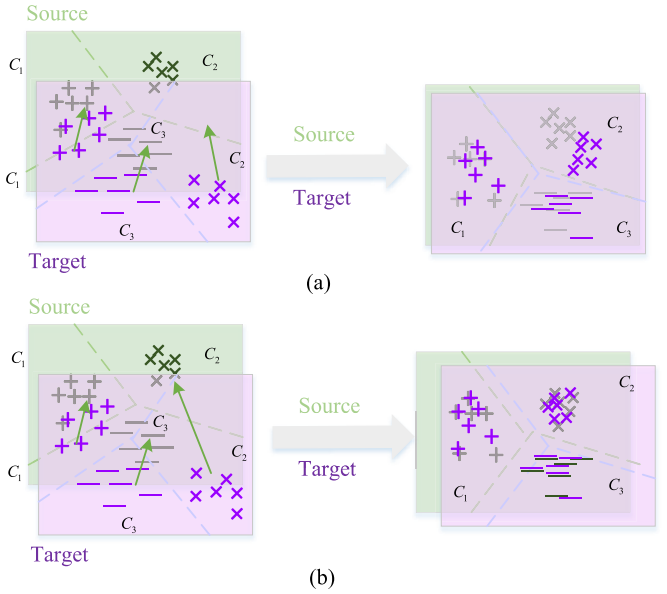


Fig. 3. Class-level domain distribution alignment using (a) conditional domain-adversarial loss and (b) adaptive domain-adversarial loss, respectively.

$$L_{\text{total}}^{t} = L_{\text{fsl}}^{t} + \lambda_{\text{adv}} L_d \tag{5}$$

where L_d denotes the adaptive conditional domain-adversarial loss from the domain discriminator, and λ_{adv} serves as the balancing term, controlling the relative contribution of the adversarial loss to the overall loss function in both domains.

Furthermore, in the testing phase, the classification process for unlabeled samples is shown in Fig. 2, where the dimension of the input data is first reduced by mapping layers, and then, spatial–spectral embedded features are extracted by a trained 3-D-CNN based on a multiscale module. (The trained feature extractor can produce discriminative features with low intraclass and high interclass variability.) Finally, the unlabeled samples are classified by the K-NN classifier, which can finally generate classification maps with labeled samples.

B. Adaptive Conditional Domain-Adversarial Strategy

A conditional domain discriminator D is employed to address the issue of domain shift between the source distribution $\mathbf{P}_s(x)$ and the target distribution $\mathbf{P}_t(x)$ in the proposed ADAFSL framework. To enhance the alignment of class-level data distributions in the source and target domains, an adaptive conditional domain-adversarial strategy is adopted, inspired by [44] and [45]. While conventional conditional domain-adversarial strategies utilize labeled data from the source domain to enhance learning in the target domain, they may face challenges when the degrees of dissimilarity in the conditional distributions of different classes between the two domains are different, leading to suboptimal adaptation performance. As depicted in Fig. 3(a), classes C_1 and C_3 can achieve basic feature alignment using the conventional conditional domain-adversarial strategy, while class C_2 may not align well due to distinct class-specific features. This is because the differences in the conditional distributions of C_1 and C_3 between the source and target domains are not significant while that of C_2 is significant, and the conventional conditional domain-adversarial strategy fails to align class C_2 between the two domains. In contrast, the proposed adaptive conditional domain-adversarial strategy introduces adaptive weights, enabling the model to iteratively focus more on the classes that are significantly different between the source and target domains. This globally aligns their condition distributions of different classes and improves the adaptation capability on the target domain, as illustrated in Fig. 3(b).

As shown in Fig. 1, the classifier module G contains discriminative information, which contains two classifiers $\textbf{Classifier}_1$ and $\textbf{Classifier}_2$, and the classifier prediction is $g = G(x)$. This can be conditioned on the adversarial adaptation of the spectral–spatial features $f = F(x)$. $\textbf{Classifier}_1$ and $\textbf{Classifier}_2$ classify and predict the features generated from F and output two class distributions $\mathbf{p}^1$ and $\mathbf{p}^2$, which are subsequently fused to obtain the final integrated prediction $\mathbf{p}$. The adaptive conditional domain-adversarial loss can be expressed as

$$\begin{aligned}\min_{D}\max_{F,G} L_d &= -E_{x_i^s \sim P_s(x)} \log\left[D\left(f_i^s, g_i^s\right)\right] \\ &\quad - E_{x_j^t \sim P_t(x)}\left(\lambda_{\text{local}} M\left(\mathbf{p}^1, \mathbf{p}^2\right) + \varepsilon\right) \log\left[1 - D\left(f_j^t, g_j^t\right)\right]\end{aligned} \tag{6}$$

where $D(f_i^s, g_i^s)$ is the probability that the discriminator predicts x to be the source-domain sample, and $1 - D(f_j^t, g_j^t)$ is the probability that the discriminator predicts x to be the target sample. $M(\mathbf{p}^1, \mathbf{p}^2)$ measures the difference between $\mathbf{p}^1$ and $\mathbf{p}^2$ by cosine distance, that is,

$$M\left(\mathbf{p}^1, \mathbf{p}^2\right) = \frac{\mathbf{p}^1 \cdot \mathbf{p}^2}{\|\mathbf{p}^1\| \|\mathbf{p}^2\|}. \tag{7}$$

The parameter λ_{local} controls the weight of adaptive adversarial loss, and ε controls the stability of the training process. In (6), when $M(\mathbf{p}^1, \mathbf{p}^2)$ is large, the feature maps of the same class in different domains do not have similar joint distributions and their semantics are not consistent. Therefore, the weights are assigned mainly to encourage G to cheat D on features that suffer from domain shift. When $M(\mathbf{p}^1, \mathbf{p}^2)$ is small, the joint distribution will have a large overlap between the source and target domains, indicating that the semantic inconsistency problem is not serious. In this case, G tends to ignore the adversarial penalty of D. As a result, the adaptive domain-adversarial strategy iteratively aligns those classes with more differences between the source and target domains.

The network of discriminator D is shown in Fig. 1, which uses stacked multilinear mappings to predict whether the input is from the source- or target-domain dataset and alleviates the domain shift problem by minimizing the classification error between the source- and target-domain data. Stacked multilinear mappings can adequately capture the multimodal structure behind complex data distributions while bringing the problem of dimensional explosion, and in this article, we use randomized multilinear mappings instead of multilinear mappings.

C. Mapping Layer and Multiscale Feature Extractor

The parameters of the mapping layer and the multiscale feature extractor are shown in Table I. The HSI cubes of the

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TABLE I
PARAMETERS OF THE MAPPING LAYER AND THE MULTISCALE FEATURE EXTRACTOR

Layer Name	Output Size	Filter Size	Padding	Stride
Input	$9 \times 9 \times b$	N/A	N	N
Mapping	$9 \times 9 \times 100$	$1 \times 1 \times 100$	N	N
Convolution1	$9 \times 9 \times 100 \times 4$	$1 \times 1 \times 1 \times 4$	N	N
Avgpool1	$9 \times 9 \times 49 \times 4$	$1 \times 1 \times 3$	N	Y
MSFE1	$9 \times 9 \times 24 \times 12$	DC:$3 \times 3 \times 3 \times 4$,R=1 DC:$3 \times 3 \times 3 \times 4$,R=1 DC:$3 \times 3 \times 3 \times 4$,R=2 AP:$1 \times 1 \times 2$	Y Y Y Y N	N N N N Y
MSFE2	$4 \times 4 \times 12 \times 36$	DC:$3 \times 3 \times 3 \times 4$,R=1 DC:$3 \times 3 \times 3 \times 4$,R=1 DC:$3 \times 3 \times 3 \times 4$,R=2 AP:$2 \times 2 \times 2$	Y Y Y Y N	N N N N Y
Convolution2	$2 \times 2 \times 10 \times 16$	$3 \times 3 \times 3 \times 16$	N	N
Avgpool2	$10 \times 1 \times 1 \times 16$	$2 \times 2 \times 1$	N	N
Flatten(Output)	160	N/A	N	N

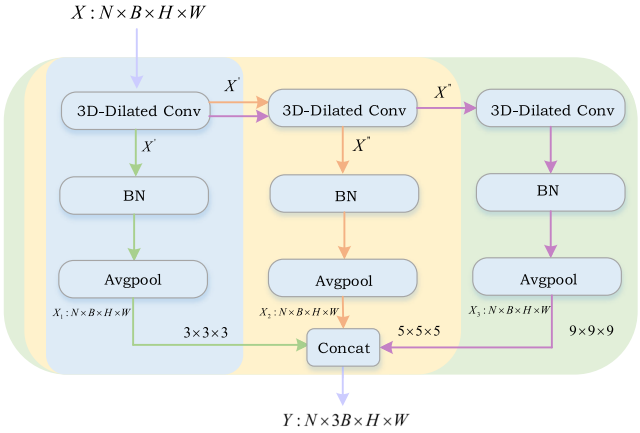


Fig. 4. Proposed MSFE module.

source and target domains are mapped to the same dimension through 3-D convolution and then fed into the multiscale feature extractor. The input of HSI cube $\mathbf{X}_h \in \mathbb{R}^{9\times9\times b}$, where 9×9 is the spatial dimension and b denotes the number of bands. The output through the mapping layer is $\mathbf{X} \in \mathbb{R}^{9\times9\times100}$. Then, 3-D CNN is used to extract the multiscale spectral–spatial features. Finally, the metric learning method is used to predict and classify the HSI dataset by FSL. Inspired by [46], the MSFE module, as shown in Fig. 4, uses cascaded dilated convolution and average pooling layer to extract multiscale spectral–spatial features from input data, which reduces the use of parameters while enlarging receptive fields. The pooling layer realizes downsampling, which accelerates the calculation speed and prevents overfitting. Assume that the input feature map is $\mathbf{X}$, then it is passed through three cascaded dilation convolution layers

$$\mathbf{X}' = \mathbf{D}_1(\mathbf{X})$$
$$\mathbf{X}'' = \mathbf{D}_2(\mathbf{X}')$$
$$\mathbf{X}''' = \mathbf{D}_3(\mathbf{X}'') \tag{8}$$

where $\mathbf{D}_i(\cdot)$ is a dilated convolution operation with a particular dilation rate. Then, they are passed through batch normalization (BN) and average pooling (Avgpool) layers

$$\mathbf{X}_1 = \mathbf{Avg}(\mathbf{BN}(\mathbf{X}'))$$
$$\mathbf{X}_2 = \mathbf{Avg}(\mathbf{BN}(\mathbf{X}''))$$
$$\mathbf{X}_3 = \mathbf{Avg}(\mathbf{BN}(\mathbf{X}''')) \tag{9}$$

where $\mathbf{BN}(\cdot)$ and $\mathbf{Avg}(\cdot)$ represent the operations of BN and Avgpool, respectively. Then, the three outputs are concatenated to form the multiscale spectral–spatial feature, that is,

$$\mathbf{Y} = [\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3]. \tag{10}$$

IV. DATASETS AND EXPERIMENTAL SETTINGS

A. Datasets

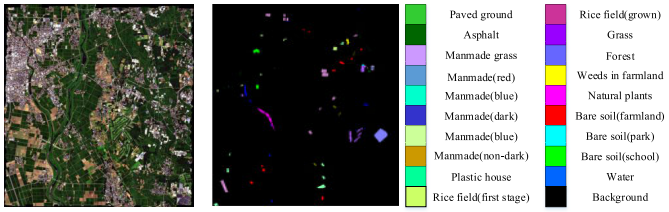
To verify the effectiveness of the proposed ADAFSL, experiments are conducted on the Chikusei dataset as the source domain and on the University of Pavia dataset, Salinas dataset, Loukia dataset, and Indian Pines (IPs) as the target domains.

1) *Chikusei Dataset*: The Chikusei dataset was taken by the Headwall Hyperspec-VNIR-C sensor in 2014 at Chikusei, Tsukishi, Japan. The dataset contains 128 bands in the wavelength range of 340–1018 nm, with a spatial size of 2517 × 2335 and a spatial resolution of 2.5 m. The dataset contains a total of 19 different classes, covering both urban and rural areas. The details of the Chikusei dataset are shown in Table II, the false-color image and ground-truth image are shown in Fig. 5.
2) *University of Pavia Dataset*: The University of Pavia (PU) dataset was acquired by the reflection optical system imaging spectrometer (ROSIS) sensor in the city of Pavia, Italy, in 2003. The Pavia University dataset

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TABLE II
CLASS NAME AND SAMPLE SIZE FOR CHIKUSEI DATASET

Class	Name	Samples
1	Water	2845
2	Bare soil(school)	2859
3	Bare soil(park)	286
4	Bare soil(farmland)	48525
5	Natural plants	4297
6	Weeds in farmland	1108
7	Forest	20516
8	Grass	6515
9	Rice field(grown)	13369
10	Rice field(first stage)	1268
11	Row crops	5961
12	Plastic house	2193
13	Manmade(non-dark)	1220
14	Manmade(dark)	7664
15	Manmade(blue)	431
16	Manmade(red)	222
17	Manmade grass	1040
18	Asphalt	801
19	Paved ground	145
	Total	77592



(a) (b)

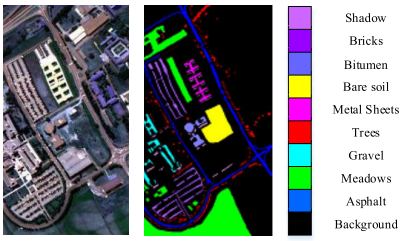
Fig. 5. Chikusei dataset: (a) false-color image and (b) ground-truth map.

TABLE III
CLASS NAME AND SAMPLE SIZE FOR THE UNIVERSITY OF PAVIA DATASET

Class	Name	Samples
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Painted metal	1345
6	Bare Soil	5029
7	Bitumen	1330
8	Self-blocking	3682
9	Shadows	947
	Total	42776

contains nine classes. The dataset contains 103 usable spectral bands after removing 12 noise bands; the spatial size of the dataset is 610 × 340, while the spatial resolution is 1.3 m. The details of the University of Pavia dataset are shown in Table III, and the false-color image and ground-truth image are shown in Fig. 6.

3) *Salinas Dataset*: The Salinas dataset was captured by the AVIRIS sensor in Salinas Valley, California. The spatial resolution of the data is 3.7 m per pixel and the size is

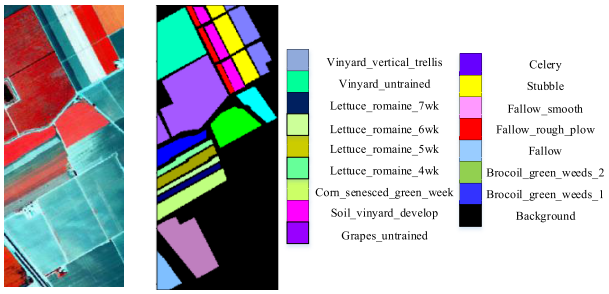


(a) (b)

Fig. 6. University of Pavia dataset: (a) false-color image and (b) ground-truth map.

TABLE IV
CLASS NAME AND SAMPLE SIZE FOR THE SALINAS DATASET

Class	Name	Samples
1	Brocoli-green-weeds-1	2009
2	Brocoli-green-weeds-2	3726
3	Fallow	1976
4	Fallow-rough-plow	1394
5	Fallow-smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes-untrained	11271
9	Soil-vinyard-develop	6203
10	Corn-senesced-green-weeds	3278
11	Lettuce-romaine-4wk	1068
12	Lettuce-romaine-5wk	1927
13	Lettuce-romaine-6wk	916
14	Lettuce-romaine-7wk	1070
15	Vinyard-untrained	7268
16	Vinyard-untrained-trellis	1807
	Total	54129



(a) (b)

Fig. 7. Salinas dataset: (a) false-color image and (b) ground-truth map.

512 × 217. The original data has 224 bands, and after removing the bands with serious water vapor absorption, 204 bands remain. The data contains 16 crop classes. The details of the Salinas dataset are shown in Table IV, and the false-color image and ground-truth image are shown in Fig. 7.

4) *IPs Dataset*: The IPs dataset was taken by the Airborne Visible/Infrared Imaging Spectrometer AVIRIS sensor in Indiana. With 224 bands from 400 to 2500 nm, 200 bands were retained after removing the spectra covering the water absorption range. The image contains 145 × 145 pixels with a spatial resolution of 20 m per pixel. The data has a total of 16 crop classes. The data

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TABLE V
CLASS NAME AND SAMPLE SIZE FOR THE IPS DATASET

Class	Name	Samples
1	Alfalfa	46
2	Corn-notil	1428
3	Corn-mintill	830
4	Corn	237
5	Grass-pasture	483
6	Grass-trees	730
7	Grass-pasture-mowed	28
8	Hay-windrowed	478
9	Oats	20
10	Soybean-notill	972
11	Soybean- mintill	2455
12	Soybean-clean	593
13	Wheat	205
14	Woods	1265
15	Buildings-Grass-Trees-Drives	386
16	Stone-Steel-Towers	93
	Total	10249

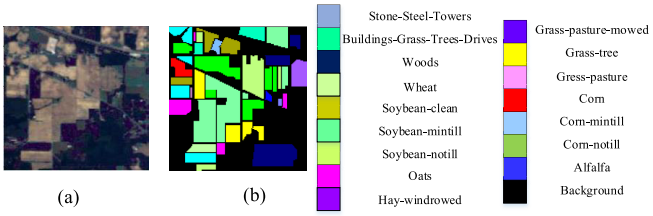


Fig. 8. IPs dataset: (a) false-color image and (b) ground-truth map.

contains 16 crop classes. The details of the IPs dataset are shown in Table V, and the false-color image and ground-truth are shown in Fig. 8.

5) *Loukia Dataset*: The Loukia dataset has been developed in the framework of the International Society for Photogrammetry and Remote Sensing (ISPRS) Scientific Initiatives. The satellite HSI collected by the Hyperion sensor (EO-1, USGS) has 176 spectral bands and is composed of 249 × 945 pixels. There are 14 consistent classes. The details of the Loukia dataset are shown in Table VI, and the false-color image and ground-truth are shown in Fig. 9.

TABLE VI
CLASS NAME AND SAMPLE SIZE FOR THE LOUKIA DATASET

Class	Name	Samples
1	Dense urban fabric	288
2	Mineral extraction sites	67
3	Non-irrigated arable land	542
4	Fruit trees	79
5	Olive groves	1401
6	Broad-leaved forest	223
7	Coniferous forest	500
8	Mixed forest	1072
9	Dense sclerophyllous vegetation	3793
10	Sparce sclerophyllous vegetation	2803
11	Sparsely vegetated areas	404
12	Rocks and sand	487
13	Water	1393
14	Coastal water	451
	Total	13503

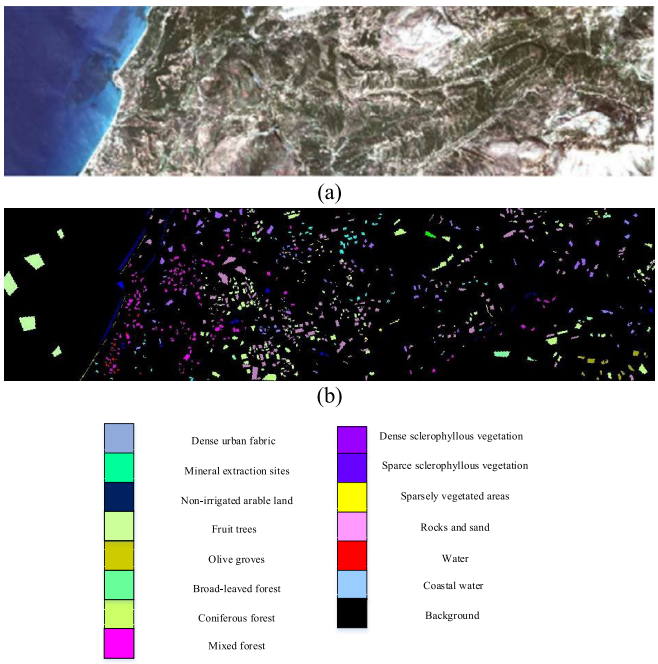


Fig. 9. Loukia dataset: (a) false-color image and (b) ground-truth map.

B. Experimental Settings

The experiments were all performed on a system with two NVIDIA Geforce RTX 2080Ti GPUs and implemented in PyTorch on Ubuntu 19.04. Referring to the experimental setup in [34] and [35], the spatial size of the input cube is chosen to be a 9 × 9 neighborhood, and convolution is used to solve the dimensional inconsistency between the source and target domains. All convolution kernels were initialized using Xavier normalization. The Adam optimizer is used for training. The number of training iterations is set to 10 000 to adequately train the network. The learning rate is set to 0.001. $\lambda_{adv} = 0.01$ and $\lambda_{local} = 1$ for the PU dataset hyperparameters, $\lambda_{adv} = 0.1$ and $\lambda_{local} = 0.01$ for the SA dataset hyperparameters, $\lambda_{adv} = 1$, $\lambda_{local} = 10$, and $\varepsilon = 0.4$ for the Ips dataset, and $\lambda_{adv} = 0.1$ and $\lambda_{local} = 0.01$ for the Loukia dataset hyperparameters. In the FSL training stage, the numbers of samples in the support and query sets are set to 1 and 19, respectively. Theoretically, the larger the sample number of the query set, the stronger the ability of the network to classify the target-domain data tends to be. We calculate the overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ) to evaluate the network performance. To avoid any bias caused by random sampling, we performed ten experiments to calculate the average results.

C. Hyperparameter Tuning

Hyperparameter sensitivity analysis was performed to evaluate the sensitivity of the proposed ADAFSL related to the two hyperparameters λ_{adv} and λ_{local}, which controls the relative importance of two kinds of losses and the adaptive weight

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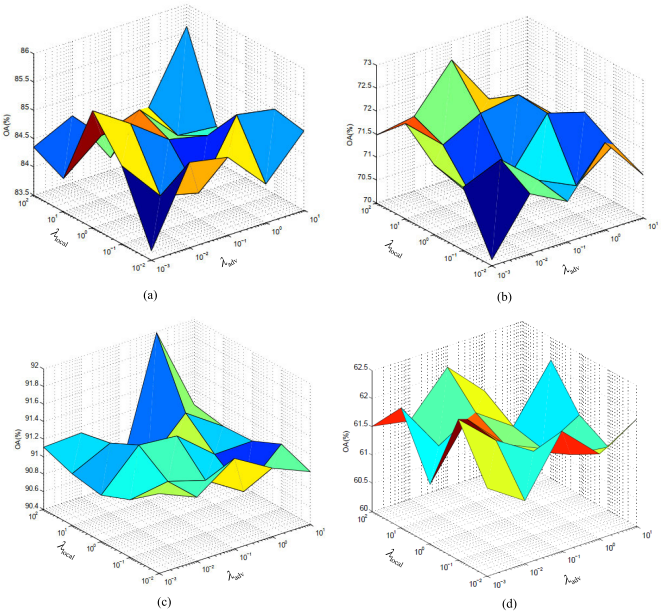


Fig. 10. Parameter tuning of λ_{local} and λ_{adv} for the proposed ADAFSL using four datasets: (a) University of Pavia dataset, (b) IPs dataset, (c) Salinas dataset, and (d) Loukia dataset.

of class-lever adversarial loss, respectively. They are selected from {0.001, 0.01, 0.1, 1, 10} to {0.01, 0.1, 1, 10, 100}, respectively. In the experimental data of different λ_{adv} and λ_{local} combinations, the trends of classification accuracy (represented by OA%) of the proposed ADAFSL on the four datasets are shown in Fig. 10. To reduce the complexity of hyperparameter optimization, we choose suboptimal parameters. For University of Pavia, Salinas, IPs, and Loukia, the suboptimal parameters are $\{\lambda_{\text{adv}} = 0.01, \lambda_{\text{local}} = 1\}$, $\{\lambda_{\text{adv}} = 0.1, \lambda_{\text{local}} = 0.01\}$, $\{\lambda_{\text{adv}} = 1, \lambda_{\text{local}} = 10\}$, and $\{\lambda_{\text{adv}} = 0.1, \lambda_{\text{local}} = 0.01\}$, respectively.

D. MSFE Module Number Selection

To verify the effect of the number of MSFE blocks in the feature extraction network on the experimental results, we performed a series of analyses using the target-domain dataset as an example, including OA, AA, κ coefficients, training time, testing time, as well as the number of parameters and the number of floating-point operations (FLOPs). The specific results are shown in Table VII. It is obvious from the table that the model shows excellent performance when the number of MSFE blocks is set to 2.

V. Experimental Results and Analysis

A. Classification Results of Different Methods

To verify the effectiveness of the proposed method in the few-shot classification task, the ADAFSL is compared with other advanced learning methods as the following.

1) SSRN [13] utilizes the residual structure to make full use of shallow features and deep features to improve the classification accuracy.
2) FDSSC [48] proposes a fast dense spectral–spatial convolution neural network framework, which uses the residual structure and dense network structure to improve the calculation efficiency while maintaining high classification accuracy.
3) DFSL [29] learns the metric space from the source dataset through a deep residual network based on 3-D-CNN. In the testing phase, SVM and K-NN methods are used to classify the unlabeled data from the test dataset.
4) DCFSL [34] uses a meta-learning paradigm for cross-domain few-shot classification to overcome domain shift using a conditional adversarial domain-adaptation strategy to achieve domain distribution alignment.
5) CMFSL [35] proposes a class-covariance metric by using the Martingale distance with task-adapted class-covariance estimation to improve the decision boundary of the nonlinear classifier by considering the distribution of each class in the potential space, thus improving the few-shot classification accuracy. In addition, the method incorporates a refinement module with spectral prior, effectively mitigating the impact of domain shift.
6) Gia-CFSL [36] proposes a graph information aggregation cross-domain FSL framework for HSI classification. The aim is to address the limitations of existing domain alignment methods, which primarily rely on local spatial information and often neglect nonlocal spatial information with strong correspondence.

For SVM, 3-D-CNN, FDSSC, and SSRN networks, five labeled samples of each class in the target domain are randomly selected to construct the training set, and the rest of the target-domain samples are selected as the test set. For DFSL-SVM, DFSL-NN, DCFSL, CMFSL, and Gia-CFSL, five labeled samples were randomly selected from each class as supervised samples in the target domain. Both the cross-domain learning method and our proposed method utilize the Chikusei dataset as the source-domain dataset. Due to the difference in spectral dimensions between the source and target domains, we employ a mapping layer to perform dimensionality reduction on both the source- and target-domain datasets. To gauge performance, we calculate OA, AA, and κ coefficient. OA is the ratio of the number of correct predictions made by the network to the overall number of pixels in all test sets, while AA is the average value of the ratio of the correct classifications to the total size of each class. The calculation of the κ coefficient is based on the confusion matrix, which is an indication of consistency. To avoid any bias induced by random sampling, we conducted ten trials and reported average results along with standard deviation.

As shown in Tables VIII–XI, the traditional SVM exhibits the lowest classification accuracy, indicating that SVM struggles to extract effective features for classification from HSIs. Moreover, when the number of labeled samples is limited, the classification accuracy of SVM is also lower than other methods, suggesting that SVM is not effective in addressing the small sample problem. In contrast, the classification methods based on CNNs show superior performance. The 3-D-CNN utilizes 3-D convolutions to extract spatial–spectral features from HSIs. The FDSSC employs a deep CNN with densely connected structures to extract spectral and spatial features, using the valid convolution method to reduce

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TABLE VII
EXPERIMENTAL RESULTS WITH DIFFERENT NUMBER OF MSFE MODULES

MSFE block Numbers		University of Pavia	Salinas	Indian Pines	Loukia
1	OA	78.42±2.77	90.99±0.90	64.31±2.16	60.30±1.54
	AA	80.38±1.42	95.19±0.70	77.07±1.29	68.03±1.07
	κ	70.02±3.18	89.90±0.99	59.70±2.41	54.77±1.66
	Training time	1735.02s	2525.56s	2479.32s	2192.61s
	Testing time	8.36s	12.51s	2.23s	2.67s
	Parameters	0.00612M	0.00612M	0.00612M	0.00612M
	FLOPs	0.164009984G	0.164009984G	0.164009984G	0.164009984G
2	OA	85.12±1.99	91.43±0.47	71.91±3.15	62.13±2.13
	AA	85.50±1.78	95.28±0.58	81.54±1.49	70.39±1.34
	κ	80.59±2.41	90.49±0.52	68.23±3.29	56.60±2.09
	Training time	1573.24s	2458.48s	3994.02s	3854.37s
	Testing time	9.19s	13.13s	2.83s	3.33s
	Parameters	0.024288M	0.024288M	0.024288M	0.024288M
	FLOPs	0.04244232G	0.04244232G	0.04244232G	0.04244232G
3	OA	82.04±1.98	89.20±0.87	69.30±3.78	61.15±3.27
	AA	86.00±1.98	93.19±1.05	81.08±1.15	69.47±1.11
	κ	77.07±2.26	87.99±0.94	65.44±4.00	55.60±3.26
	Training time	3483.57s	9454.35s	9652.75s	4545.10s
	Testing time	15.02s	20.39s	6.91s	5.40s
	Parameters	0.125448M	0.125448M	0.125448M	0.125448M
	FLOPs	33.875051008G	33.875051008G	33.875051008G	33.875051008G

TABLE VIII
CLASSIFICATION RESULTS (%) FOR THE UNIVERSITY OF PAVIAU DATASET (FIVE LABELED SAMPLES PER CLASS)

	SVM [31]	3D-CNN [47]	FDSSC [48]	SSRN [13]	DFSL-NN [29]	DFSL-SVM [29]	DCFSL [34]	CMFSL [35]	Gia-CFSL [36]	ADAFSL
1	53.32	70.12	76.10	81.03	90.13	73.41	80.37	77.31	88.12	87.80
2	78.14	62.43	59.57	65.48	83.02	82.85	85.49	85.54	85.95	86.92
3	44.34	50.78	69.01	61.16	67.34	66.27	63.74	60.15	66.44	59.96
4	90.08	84.29	65.48	98.43	95.86	96.19	94.27	96.29	92.78	88.58
5	91.81	85.60	98.18	99.98	99.38	99.03	99.51	99.68	99.18	99.36
6	10.65	49.37	77.37	73.21	66.08	75.28	76.30	77.22	75.03	78.47
7	95.64	70.13	90.32	94.67	77.28	80.91	79.42	76.74	74.58	86.20
8	52.03	74.66	69.82	96.73	61.09	65.52	59.94	74.91	72.12	82.26
9	99.92	94.56	99.40	99.92	99.50	99.93	98.67	99.92	94.67	99.89
OA	64.74±1.79	65.81±1.17	68.97±3.30	76.78±2.01	81.09±1.73	79.97±1.30	81.52±2.72	82.39±1.00	83.60±2.29	**85.12±1.99**
AA	68.37±2.08	71.33±2.40	78.36±4.69	86.59±2.09	82.19±1.06	82.15±1.42	81.97±1.23	83.09±1.46	83.21±1.86	**85.50±1.78**
κ	54.28±2.12	57.27±1.71	61.53±4.04	71.01±2.52	74.02±1.85	74.13±1.39	76.10±2.71	77.21±1.76	78.67±2.73	**80.59±2.41**

high dimensionality. The SSRN incorporates a single-branch cascade of traditional 3-D convolutions, including residual connections to form a residual network. These deep-learning methods outperform the traditional SVM approach as they can extract distinct and reliable features through deep representations, and the appropriate multilayer architectures facilitate the effective classification of hyperspectral datasets. However, under the constraint of limited labeled samples, these methods fail to achieve promising classification accuracy.

Compared with deep-learning methods, methods based on deep FSL, including DFSL-SVM and DFSL-NN, can achieve better classification performance than CNN-based methods in the absence of labeled samples. Meanwhile, cross-domain FSL methods can achieve better classification accuracy than deep FSL methods. DCFSL, as a cross-domain FSL method, can learn transferable knowledge from the source-domain dataset to help classify unknown samples in the target domain. At the same time, the domain-adaptive module is introduced into the DCFSL method, which can reduce the influence of domain shift caused by cross-domain and obtain better classification results. On the basis of DCFSL, CMFSL proposes a kind of class-covariance measure learning metric space, which improves the decision boundary of the nonlinear classifier by considering the distribution of each class in the potential space, thus improving the classification accuracy of a few shots. The Gia-FSL method, on the other hand, mitigates the impact of domain shift on FSL by considering nonlocal spatial information with strong correspondence when performing domain alignment and by using feature-level and distribution-level cross-domain graph alignments.

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TABLE IX
CLASSIFICATION RESULTS (%) FOR THE SALINAS DATASET (FIVE LABELED SAMPLES PER CLASS)

	SVM [31]	3D-CNN [47]	FDSSC [48]	SSRN [13]	DFSL-NN [29]	DFSL-SVM [29]	DCFSL [34]	CMFSL [35]	Gia-CFSL [36]	ADAFSL
1	96.27	98.24	97.72	99.92	95.00	99.98	98.57	98.53	98.95	99.19
2	99.84	99.81	95.48	99.63	89.73	99.87	99.44	99.77	98.63	99.68
3	65.21	73.21	94.29	93.04	91.61	95.18	98.15	97.87	93.06	97.10
4	97.31	97.92	98.99	98.48	98.23	99.50	98.63	98.80	98.67	99.57
5	97.84	96.46	96.14	94.32	97.25	94.47	94.44	93.64	87.64	95.74
6	99.66	99.84	98.68	99.99	99.89	98.12	98.72	99.62	99.16	99.26
7	99.48	99.43	99.39	99.13	99.72	99.58	99.29	99.61	98.37	99.30
8	37.86	58.61	62.22	80.08	58.34	65.36	75.21	82.08	74.21	77.67
9	95.79	97.65	98.45	96.34	99.14	99.90	99.67	85.53	98.13	99.66
10	70.13	61.13	63.06	91.74	70.59	87.11	85.95	80.49	81.82	84.64
11	92.29	91.99	93.74	63.45	96.96	99.44	97.98	98.30	97.43	99.08
12	79.91	87.83	82.27	98.67	86.07	99.64	99.63	99.92	98.71	99.92
13	99.54	99.56	99.89	99.84	99.98	99.23	99.12	99.31	98.63	99.60
14	98.07	96.91	96.06	99.63	97.80	99.39	98.94	96.90	97.89	99.44
15	84.91	69.94	65.30	54.62	78.18	70.79	67.90	65.54	74.24	84.40
16	97.11	94.92	97.84	97.82	98.51	95.27	95.12	89.88	87.43	90.19
OA	79.98±1.95	82.56±0.54	83.19±1.67	84.52±2.02	85.14±1.16	87.05±1.14	88.77±0.39	89.30±0.62	88.07±2.30	**91.43±0.47**
AA	88.17±1.29	88.99±1.25	89.97±1.34	91.67±1.41	91.69±1.46	93.61±2.52	94.15±0.57	93.69±0.71	92.69±1.46	**95.28±0.58**
κ	78.00±1.08	80.66±1.64	81.31±1.81	82.90±2.82	83.51±1.28	90.64±1.46	87.52±0.39	88.09±0.69	86.75±2.53	**90.49±0.52**

TABLE X
CLASSIFICATION RESULTS (%) FOR THE IPS DATASET (FIVE LABELED SAMPLES PER CLASS)

	SVM [31]	3D-CNN [47]	FDSSC [48]	SSRN [13]	DFSL-NN [29]	DFSL-SVM [29]	DCFSL [34]	CMFSL [35]	Gia-CFSL [36]	ADAFSL
1	81.34	86.63	89.86	95.35	37.00	93.49	95.12	100.00	92.68	100.00
2	33.40	28.60	37.47	41.74	46.78	39.96	32.40	38.08	46.99	51.03
3	39.54	55.92	54.46	53.9	39.90	148.04	53.66	54.52	45.94	64.10
4	76.80	61.85	70.52	51.28	28.08	87.36	72.56	83.21	76.85	81.21
5	29.95	45.98	86.88	21.61	77.88	69.18	73.29	82.22	72.76	79.75
6	44.58	82.60	79.13	86.76	94.13	90.71	90.16	86.14	80.73	92.63
7	61.96	94.41	98.26	100.00	30.81	100.00	100.00	99.46	99.57	100.00
8	40.30	71.31	82.15	77.69	98.04	86.26	90.20	93.90	82.92	92.30
9	98.33	100.00	100.00	100.00	19.36	100.00	100.00	98.33	100.00	100.00
10	40.41	39.29	53.80	55.83	58.26	61.97	63.15	58.92	60.85	59.54
11	27.62	51.75	47.82	58.29	74.33	54.04	66.72	68.87	62.19	69.93
12	29.44	24.37	21.14	22.16	34.27	37.13	40.13	34.48	47.18	59.93
13	96.54	97.18	95.02	96.78	75.00	99.17	99.33	99.94	97.50	97.20
14	80.58	78.66	68.05	81.89	94.21	92.48	84.65	86.10	78.21	87.86
15	59.70	60.02	90.72	84.20	55.58	62.82	70.43	80.22	67.30	71.50
16	99.72	98.41	99.55	100.00	86.99	99.62	98.11	98.86	98.64	97.73
OA	43.95±2.20	54.42±1.72	57.98±1.12	59.69±1.24	60.18±4.63	63.65±1.86	65.70±1.42	67.48±1.95	64.31±2.97	**71.91±3.15**
AA	58.75±4.58	67.31±4.14	73.43±0.93	70.47±1.87	59.41±1.96	76.39±1.81	76.87±1.58	78.96±1.61	75.64±1.92	**81.54±1.49**
κ	37.74±1.97	50.37±3.83	52.87±1.08	54.52±1.63	55.66±4.70	59.28±1.90	61.18±1.32	63.15±1.24	59.76±3.21	**68.23±3.39**

Our method is also a cross-domain FSL method, which uses a multiscale feature extraction network to learn the metric space and an adaptive conditional domain-adversarial strategy to mitigate the effect of cross-domain domain shift. When evaluated on the University of Pavia dataset in a comparative experiment, our method demonstrated an improvement of 1.52% in OA, 2.29% in AA, and 1.92% in the Kappa coefficient compared to the optimal Gia-CFSL method. In addition, on the IPs, Salinas, and Loukia datasets, our method achieved increases of 4.43%, 2.13%, and 2.71% in OA and 2.58%, 1.59%, and 0.13% in AA compared to the optimal CMFSL method, while the Kappa coefficients saw improvements of 5.08%, 2.40%, and 2.54%, respectively. This shows the effectiveness of the proposed ADAFSL and proves its superiority in cross-domain HSI classification.

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TABLE XI
CLASSIFICATION RESULTS (%) FOR THE LOUKIA DATASET (FIVE LABELED SAMPLES PER CLASS)

	SVM [31]	3D-CNN [47]	FDSSC [48]	SSRN [13]	DFSL-NN [29]	DFSL-SVM [29]	DCFSL [34]	CMFSL [35]	Gia-CFSL [36]	ADAFSL
1	57.29	54.42	36.28	55.69	60.49	60.35	65.87	70.50	57.14	64.31
2	100.00	91.92	100.00	100.00	95.81	95.81	92.26	97.58	96.61	94.19
3	34.04	42.33	52.70	50.05	75.57	73.41	73.74	74.67	72.12	77.69
4	53.18	69.05	76.58	79.49	70.81	62.16	69.19	73.20	64.73	74.86
5	35.64	22.67	22.28	40.69	44.03	45.87	41.76	52.60	45.95	56.93
6	51.20	43.90	70.03	61.39	61.38	61.19	59.54	63.15	54.08	54.22
7	59.60	65.21	61.55	62.74	53.74	54.26	59.19	60.00	57.15	65.41
8	77.47	71.53	62.58	76.44	59.91	59.29	64.59	64.84	56.58	57.24
9	28.97	34.65	33.69	34.00	30.47	29.81	42.03	37.83	42.00	50.61
10	40.09	42.61	64.07	44.92	58.66	57.88	53.04	54.72	52.99	51.14
11	68.88	70.65	52.80	61.44	64.31	65.31	61.05	60.74	66.27	69.17
12	84.01	81.55	62.52	86.25	86.47	86.56	78.76	75.76	81.12	68.13
13	100.00	100.00	100.00	100.00	99.65	100.00	99.86	100.00	99.63	100.00
14	99.93	99.93	70.26	100.00	99.91	99.96	100.00	99.93	99.84	99.69
OA	51.47±1.84	52.10±1.66	53.77±1.61	55.20±2.23	56.88±0.74	56.62±0.56	58.94±1.31	59.42±1.61	58.51±3.11	**62.13±2.13**
AA	63.59±1.58	63.47±1.85	61.81±2.71	68.08±1.12	68.66±1.48	67.99±1.96	68.63±1.05	70.26±0.81	67.59±1.82	**70.39±1.34**
κ	45.60±1.93	46.30±1.49	47.65±2.26	49.86±2.25	51.41±0.81	51.16±0.61	53.36±1.34	54.06±1.57	52.78±3.20	**56.60±2.09**

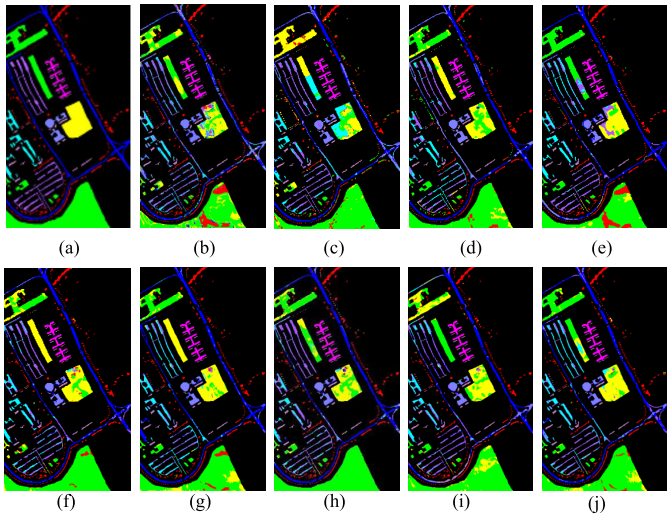


Fig. 11. Classification maps obtained via various algorithms for the University of PaviaU dataset: (a) ground-truth map, (b) 3-D-CNN (65.81%), (c) FDSSC (68.97%), (d) SSRN (76.78%), (e) DFSL-SVM (81.09%), (f) DFSL-NN (79.97%), (g) DCFSL (81.52%), (h) CMFSL (82.39%), (i) Gia-CFSL (83.60%), and (j) ADAFSL (85.12%).

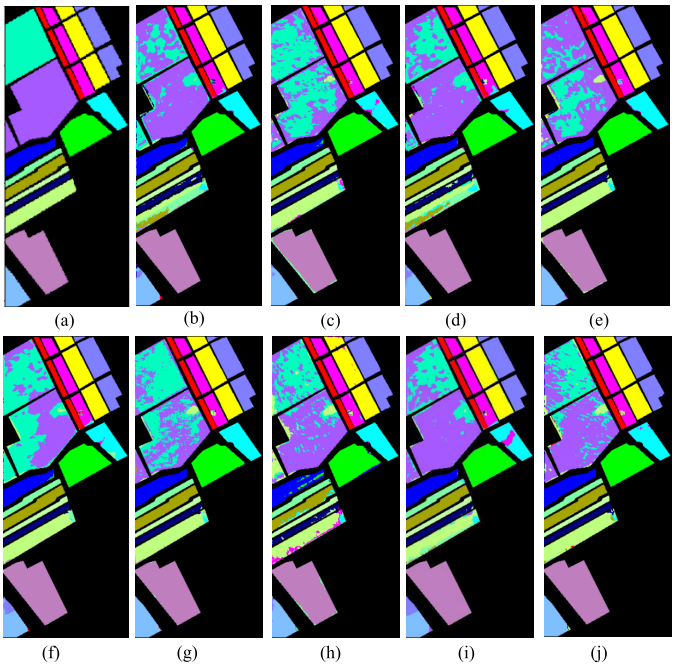


Fig. 12. Classification maps obtained via various algorithms for the Salinas dataset: (a) ground-truth map, (b) 3-D-CNN (82.56%), (c) FDSSC (83.19%), (d) SSRN (84.52%), (e) DFSL-SVM (85.14%), (f) DFSL-NN (87.09%), (g) DCFSL (88.77%), (h) CMFSL (89.30%), (i) Gia-CFSL (88.07%), and (j) ADAFSL (91.43%).

B. Visual Analysis

For visual comparison, Figs. 11–14 show the classification maps of the various methods on four datasets, where the maps produced by the proposed ADAFSL match the ground-truth map best, further demonstrating the superiority of ADAFSL. The classification maps generated by SVM and three CNN-based methods (3-D-CNN, FDSSC, and SSRN) contain substantial noise, while the FSL-based methods (DFSL-SVM, DFSL-NN, DCFSL, CMFSL, Gia-CFSL, and the proposed ADAFSL) produce smoother results, although certain classes show evident misclassifications. ADAFSL provides the most accurate and smooth classification result plots on the four target datasets compared to other FSL-based methods. Taking the IPs dataset as an example, due to its high noise level, numerous classes, and uneven sample distribution, handling spectral uncertainty is more challenging compared to other datasets. This has an impact on the performance of many classification networks of this type. However, in contrast, the suggested ADAFSL method demonstrates significant advantages in tackling such issues, resulting in impressive classification performance compared to other methods.

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TABLE XII
ABLATION EXPERIMENT

		University of Pavia	Salinas	Indian Pines	Loukia
Net A	OA	79.59±2.82	89.34±0.83	70.86±1.91	60.02±1.76
	AA	84.00±2.84	93.70±0.36	81.06±0.87	67.59±1.75
	κ	74.22±2.71	88.16±0.92	67.16±2.07	54.45±1.47
	Parameters	0.024288M	0.024288M	0.024288M	0.024288M
	FLOPs	0.04244232G	0.04244232G	0.04244232G	0.04244232G
Net B	OA	78.88±2.13	90.26±0.17	62.70±1.97	60.12±1.76
	AA	80.86±1.62	94.91±0.59	76.36±1.41	70.23±1.35
	κ	72.99±2.69	89.17±0.21	58.05±1.96	54.73±1.69
	Parameters	0.034920M	0.034920M	0.034920M	0.034920M
	FLOPs	0.059785344G	0.059785344G	0.059785344G	0.059785344G
Net C	OA	85.12±1.99	91.43±0.47	71.91±3.15	62.13±2.13
	AA	85.50±1.78	95.28±0.58	81.54±1.49	70.39±1.34
	κ	80.59±2.41	90.49±0.52	68.23±3.29	56.60±2.09
	Parameters	0.024288M	0.024288M	0.024288M	0.024288M
	FLOPs	0.04244232G	0.04244232G	0.04244232G	0.04244232G

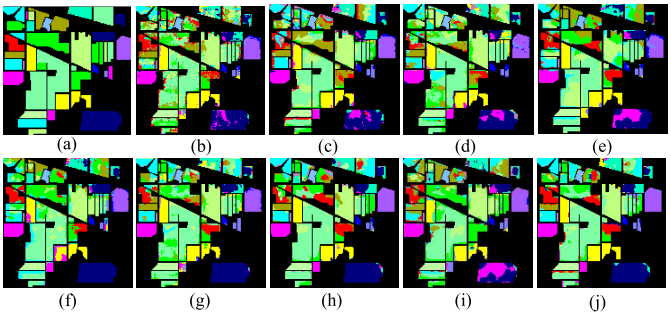


Fig. 13. Classification maps obtained via various algorithms for the IPs dataset: (a) ground-truth map, (b) 3-D-CNN (54.42%), (c) FDSSC (57.98%), (d) SSRN (59.69%), (e) DFSL-SVM (60.18%), (f) DFSL-NN (63.65%), (g) DCFSL (65.70%), (h) CMFSL (67.48%), (i) Gia-CFSL (64.31%), and (j) ADAFSL (71.91%).

C. Ablation Experiments

A battery of ablation experiments is conducted to verify various facets of the ADAFSL design. In the proposed network, feature reuse is realized by using a feature cascade, and different levels of spatial–spectral features are extracted by the MSFE module. The few-shot loss function of two classifiers' decision fusion is used to adjust the FSL performance of the source domain and target domain. In the domain-adaptation strategy, the adaptive conditional domain-adversarial loss function is used to align data distribution to reduce the influence of domain shift. We design two additional networks to compare with the proposed ADAFSL, as given in Table XII, to gauge the effects of two main components within ADAFSL. These networks are as follows. Variations on our proposed ADAFSL framework.

1) Network A: ADAFSL network without the MSFE module.
2) Network B: ADAFSL network without the domain adaptation strategy.
3) Network C: The proposed ADAFSL network.

The resulting ablation networks are evaluated in terms of OA, AA, κ, parameters, and FLOPs in Table XII. The proposed ADAFSL outperforms both of the other two networks, demonstrating that the multiscale spatial–spectral extraction module and domain-adaptation strategy act complementarily and both contribute to the classification performance of the proposed ADAFSL network.

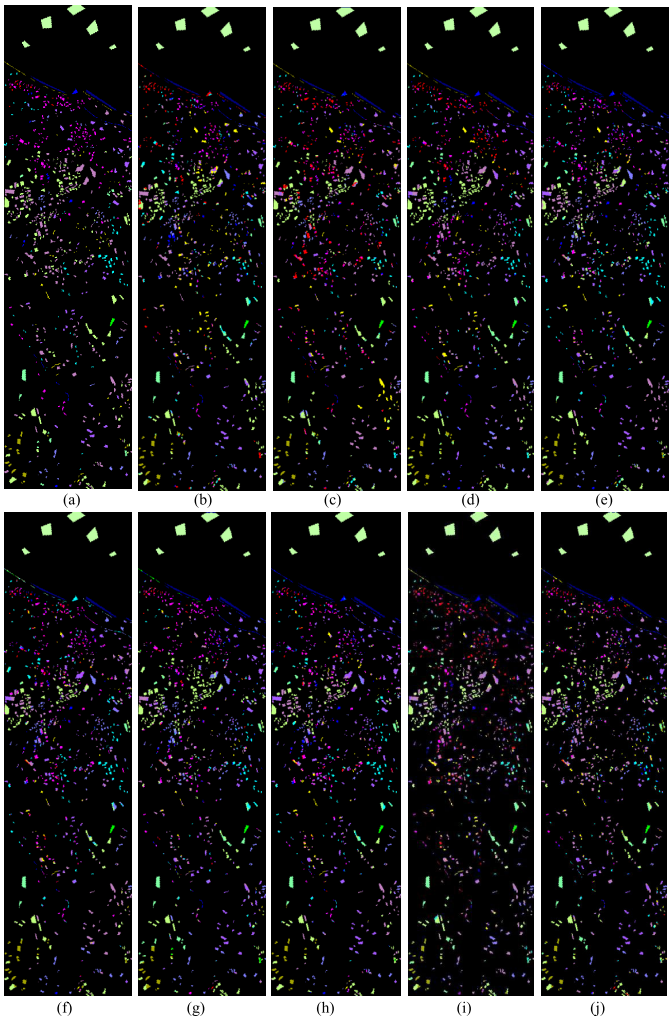


Fig. 14. Classification maps obtained via various algorithms for the Loukia dataset: (a) ground-truth map, (b) 3-D-CNN (52.10%), (c) FDSSC (53.77%), (d) SSRN (55.20%), (e) DFSL-SVM (56.88%), (f) DFSL-NN (56.62%), (g) DCFSL (58.94%), (h) CMFSL (59.42%), (i) Gia-CFSL (58.51%), and (j) ADAFSL (62.13%).

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TABLE XIII
COMPUTATIONAL TIME (SECONDS), FLOPS, AND PARAMETERS ON EACH DATASET WITH DIFFERENT METHODS

	Methods	DFSL-SVM	DFSL-NN	DCFSL	CMFSL	Gia-CFSL	ADAFSL
University of Pavia	Training time	1896.71s	834.76s	1482.50s	991.87s	2245.34s	1573.24s
	Testing time	6.25s	11.12s	6.23s	5.82s	4.37s	9.19s
	FLOPs	0.041135921G	0.041135921G	0.059785344G	0.244174752G	5.332537344G	0.04244232G
	#params	0.038140M	0.038140M	0.034920M	0.613976M	0.185192M	0.024288M
Salinas	Training time	2695.96s	1315.05s	2283.42s	1696.61s	5937.97s	2458.48s
	Testing time	2.26s	11.45s	7.79s	7.34s	6.33s	13.13s
	FLOPs	0.041135921G	0.041135921G	0.059785344G	0.244174752G	5.332537344G	0.04244232G
	#params	0.038140M	0.038140M	0.034920M	0.613976M	0.185192M	0.024288M
Indian Pines	Training time	3006.98s	1312.27s	2259.96s	1901.68s	3398.78s	3994.02s
	Testing time	1.57s	2.34s	1.68s	1.47s	1.15s	2.83s
	FLOPs	0.041135921G	0.041135921G	0.059785344G	0.244174752G	5.332537344G	0.04244232G
	#params	0.038140M	0.038140M	0.034920M	0.613976M	0.185192M	0.024288M
Loukia	Training time	1575.18s	1588.36s	1255.28s	1468.06s	2986.22s	3854.37s
	Testing time	1.74s	3.08s	2.53s	1.51s	1.44s	3.33s
	FLOPs	0.041135921G	0.041135921G	0.059785344G	0.244174752G	5.332537344G	0.04244232G
	#params	0.038140M	0.038140M	0.034920M	0.613976M	0.185192M	0.024288M

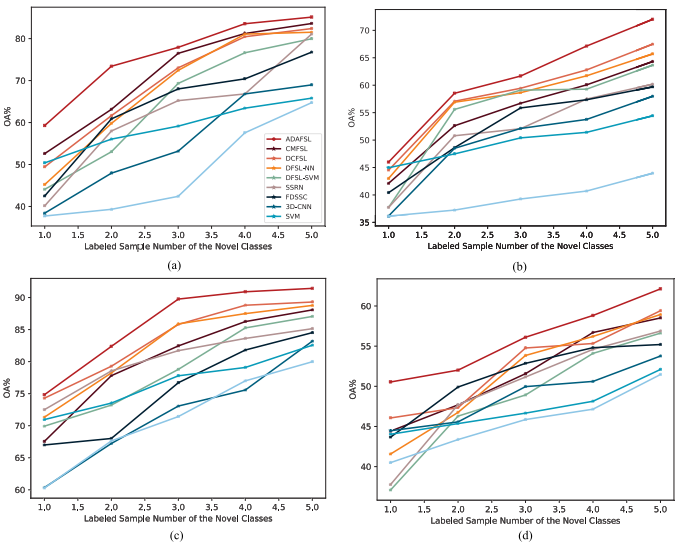


Fig. 15. Classification accuracy (OA %) with different numbers of labeled samples on target datasets. (a) University of Pavia dataset, (b) IPs dataset, (c) Salinas dataset, and (d) Loukia dataset.

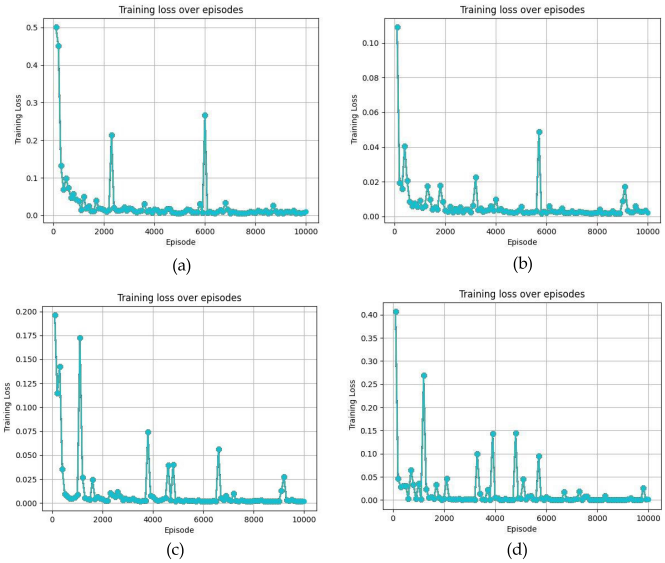


Fig. 16. Total loss iteration curve. (a) University of Pavia dataset, (b) IPs dataset, (c) Salinas dataset, and (d) Loukia dataset.

D. Classification Accuracies With Different Numbers of Labeled Samples

To analyze the influence of different numbers of labeled training samples on the classification performance of SVM, CNN-based methods, and several FSL-based methods, we randomly select the number of labeled samples per class from {1, 2, 3, 4, 5}, and the final classification results for four datasets are reported in Fig. 15. The classification performance of the various methods decreases with a decreasing number of training samples, which is as expected. However, the proposed ADAFSL always produces higher classification accuracy when compared to the other methods, even with an exceedingly small training sample size (e.g., only one training sample per class).

E. Computational Complexity

To demonstrate the computational efficiency of different FSL methods, the training, testing time, and the number of parameters and FLOPs are analyzed, with five labeled samples per class in the target-domain dataset, as shown in Table XIII. For cross-domain methods (DCFSL, CMFSL, Gia-CFSL, and our ADAFSL), the training time is mainly consumed by transfer learning, and the testing time is mainly the predicting time of unlabeled samples for each HSI dataset. The number of FLOPs and parameters was investigated to evaluate the computational complexity of deep-learning methods. From Table XIII, it can be observed that, compared with other methods, the computational time, FLOPs, and the number of parameters are all acceptable. All experiments are conducted on a system with two NVIDIA Geforce RTX 2080Ti GPUs. Implementation is in PyTorch on Ubuntu 19.04. The memory is 64 GB.

F. Training Loss Analysis

To study the variation of the total loss function of the proposed ADAFSL method with the number of training episodes, we experimentally plot the trend of loss variation on different datasets. In Fig. 16, we can clearly observe that despite some

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fluctuations in the model training loss plot (noise effects in the public dataset/unseen classes in the target dataset), our proposed model is still able to reduce the loss value quickly through learning, and the overall trend of the total loss function gradually decreases and stabilizes during the training process. This trend shows that our ADAFSL method exhibits good convergence on different datasets, and it is still able to learn effectively and improve performance even when facing some challenging situations, demonstrating the robustness and applicability of our method.

VI. Conclusion

In this article, a novel ADAFSL method is proposed to solve the problem of HSI cross-domain classification in the case of sufficient labeled samples in the source domain and few labeled samples in the target domain. Specifically, an adaptive conditional-adversarial strategy is adopted by measuring the distance between labeled and unlabeled data by Euclidean and class-covariance metrics, which are used to produce a local alignment score map by comparing their prediction maps, thus assigning different weights to different conditional domain adversarial losses. In addition, a multiscale spatial–spectral feature extraction module based on feature cascade is proposed to obtain more discriminant embedding features and enhance the adaptive conditional domain alignments. The experimental results for target-domain datasets show that the proposed ADAFSL is superior to other advanced methods. Despite obtaining promising results, more studies are required to further improve the performance of the ADAFSL method. In future work, we will try to investigate lightweight methods that are conducive to deploying them in resource-constrained situations (devices).

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